WORKED SOLUTION

STEP II 2024, Question 7

Pure • Curve Sketching

STEP II 2024, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Each bracket is the equation of a circle, so C_1 is just two circles. For C_2, multiplying the brackets out collapses everything into x^2+y^2 and x^2, so cutting with a horizontal line gives a quadratic in x^2. Counting intersections then comes down to counting the positive roots of that quadratic, which the discriminant, the sum and the product of the roots settle between them.

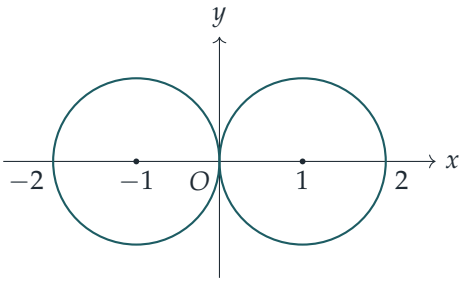
Solution

Throughout, write

$$A = y^2+(x-1)^2-1 = x^2+y^2-2x, \qquad B = y^2+(x+1)^2-1 = x^2+y^2+2x.$$

Part (i)

A product is zero exactly when one of its factors is zero, so C_1 is the union of the circles $A=0$ and $B=0$. These are the circles of radius 1 centred at $(1,0)$ and $(-1,0)$. Both pass through the origin, where they touch (their common tangent there is the y-axis). The highest points are $(\pm 1, 1)$ and the curve crosses the x-axis at $x=-2, 0, 2$.



Part (ii)(a)

Since A and B have the form $s - 2x$ and $s + 2x$ with $s = x^2 + y^2$,

$$AB = (x^2 + y^2)^2 - 4x^2.$$

On the line $y = k$ the equation $AB = \frac{1}{16}$ becomes

$$\begin{aligned}(x^2 + k^2)^2 - 4x^2 &= \tfrac{1}{16}\\ x^4 + 2k^2x^2 + k^4 - 4x^2 - \tfrac{1}{16} &= 0\\ x^4 + 2(k^2 - 2)x^2 + \left(k^4 - \tfrac{1}{16}\right) &= 0,\end{aligned}$$

as required.

Now put $u = x^2$, giving

$$u^2 + 2(k^2 - 2)u + \left(k^4 - \tfrac{1}{16}\right) = 0. \qquad (*)$$

Each root $u > 0$ gives two intersection points $x = \pm\sqrt{u}$, a root $u = 0$ gives one point ($x = 0$), and a root $u < 0$ gives none. So we need to know the signs of the roots of $(*)$. Three quantities tell us:

- the quarter-discriminant $(k^2 - 2)^2 - \left(k^4 - \frac{1}{16}\right) = \frac{65}{16} - 4k^2$, which is positive, zero or negative as k is less than, equal to or greater than $\frac{\sqrt{65}}{8}$;
- the sum of the roots, $2(2 - k^2)$;
- the product of the roots, $k^4 - \frac{1}{16}$, which has the sign of $k - \frac{1}{2}$ for $k > 0$.

Note that $\frac{1}{2} < \frac{\sqrt{65}}{8} \approx 1.008$, and that whenever the roots are real we have $k^2 \leq \frac{65}{64} < 2$, so their sum is positive.

> **Ask yourself** If the product of the roots is negative, do we need to check the discriminant?

No: a negative product forces the roots to be real (the discriminant $b^2 - 4ac$ is then positive) and of opposite signs. Now we take the cases in turn.

- $0 < k < \frac{1}{2}$: the product is negative, so there is exactly one positive root. **2 points.**
- $k = \frac{1}{2}$: $(*)$ becomes $u^2 - \frac{7}{2}u = 0$, so $u = 0$ or $u = \frac{7}{2}$, giving $x = 0$ and $x = \pm\sqrt{7/2}$. **3 points.**
- $\frac{1}{2} < k < \frac{\sqrt{65}}{8}$: the discriminant is positive, so there are two distinct real roots. Their product and sum are both positive, so both roots are positive. **4 points.**
- $k = \frac{\sqrt{65}}{8}$: there is a repeated root $u = 2 - k^2 = \frac{63}{64} > 0$, giving $x = \pm\frac{\sqrt{63}}{8}$. **2 points.**
- $k > \frac{\sqrt{65}}{8}$: no real roots. **No points.**

Part (ii)(b)

By (a), the line $y = k$ meets C_2 when $0 < k \leq \frac{\sqrt{65}}{8}$ and misses it for larger k. The equation of C_2 involves y only through y^2, so negative k gives nothing higher. The greatest y-coordinate on C_2 is therefore $\frac{\sqrt{65}}{8}$, attained at $x = \pm\frac{\sqrt{63}}{8}$.

On C_1 the highest points are the tops of the circles, $(\pm 1, 1)$. Since $\frac{\sqrt{63}}{8} < \frac{\sqrt{64}}{8} = 1$, the highest points of C_2 are **closer to the y-axis** than those of C_1 (though slightly higher, as $\frac{\sqrt{65}}{8} > 1$).

Part (ii)(c)

Suppose $A < 0$ and $B < 0$. From $A < 0$ we get $2x > x^2 + y^2 \geq 0$, so $x > 0$. From $B < 0$ we get $-2x > x^2 + y^2 \geq 0$, so $x < 0$. These contradict each other, so A and B cannot both be negative. (Geometrically, the two open discs don't overlap.)

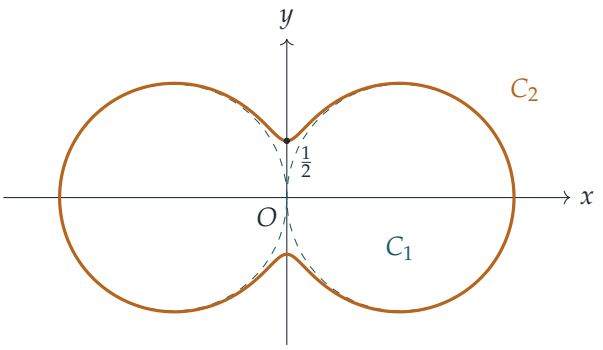
On C_2 we have $AB = \frac{1}{16} > 0$, so A and B are nonzero with the same sign. They can't both be negative, so both are positive. Now $A > 0$ says that $(x-1)^2 + y^2 > 1$, so the point lies strictly outside the circle centred at $(1, 0)$. Likewise $B > 0$ puts it strictly outside the circle centred at $(-1, 0)$. Hence every point of C_2 lies outside both circles: C_2 lies entirely outside C_1.

Part (ii)(d)

Useful features of C_2 for the sketch:

- It is symmetric in both axes, since only x^2 and y^2 appear.
- On $y = 0$, $(*)$ gives $u = 2 + \frac{\sqrt{65}}{4}$, so C_2 crosses the x-axis at $x = \pm\sqrt{2 + \frac{\sqrt{65}}{4}} \approx \pm 2.004$, just beyond ± 2.
- On $x = 0$, $k^4 = \frac{1}{16}$, so C_2 crosses the y-axis at $(0, \pm\frac{1}{2})$.
- Its highest and lowest points are $\left(\pm\frac{\sqrt{63}}{8}, \pm\frac{\sqrt{65}}{8}\right) \approx (\pm 0.99, \pm 1.01)$.

The counts in (a) show the shape. For $|y| < \frac{1}{2}$ there are just two points (the outer sides). For $\frac{1}{2} < |y| < \frac{\sqrt{65}}{8}$ there are four, so the curve dips down to $(0, \pm\frac{1}{2})$ between two humps. C_2 is therefore a single closed "peanut" enclosing both circles, pinched in at the y-axis.



(Writing C_2 in polar form gives $r^2 = 2\cos^2\theta + \sqrt{4\cos^4\theta + \frac{1}{16}}$, which is how the curve above was drawn.)

Checks

- At $k = \frac{\sqrt{65}}{8}$ the discriminant $\frac{65}{16} - 4k^2$ vanishes, matching the top of the curve. The repeated root $u = \frac{63}{64}$ gives $x^2 + y^2 = \frac{128}{64} = 2$ and $4x^2 = \frac{63}{16}$, and indeed $2^2 - \frac{63}{16} = \frac{1}{16}$.
- The point $(0, \frac{1}{2})$: $A = B = \frac{1}{4}$, so $AB = \frac{1}{16}$.
- Counting the crossings of $y = k$ with C_2 by computer for $k = 0.1$, 0.49, 0.51, 1.0, 1.007 and 1.01 gives 2, 2, 4, 4, 4 and 0 points respectively, and a numerical search finds the maximum height $1.00778 \approx \frac{\sqrt{65}}{8}$ at $x \approx 0.9922 \approx \frac{\sqrt{63}}{8}$.

Common slips

- Treating $u = 0$ as giving two points: $x = \pm\sqrt{0}$ is a single point, which is why $k = \frac{1}{2}$ gives 3 intersections, not 4.
- Forgetting the tangency case $k = \frac{\sqrt{65}}{8}$ in the count.
- In (c), concluding only "A and B have the same sign" without ruling out both negative, which is the whole point of the first half.

Beyond STEP

C_2 belongs to the family $(x^2 + y^2)^2 - 4x^2 = c$, the level curves of the function AB. As c increases the curve changes shape: two separate ovals inside the circles (for $c < 0$), then the figure-of-eight C_1 at $c = 0$, then a single peanut, whose dimples disappear once $c \geq 4$. The change of topology happens exactly at the level of the critical point of AB at the origin. Following how level curves change as they pass critical points is the starting idea of Morse theory, and the closely related Cassini ovals, with the lemniscate of Bernoulli as their figure-of-eight, behave in just the same way. Siklos, *Advanced*

Problems in Mathematics, has more STEP-style curve sketching of implicit curves like this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.